Preparing Wet Mount Microscope Slides

Preparing a specimen

Light microscopes function by allowing light to pass through the specimen to our eye. If the specimen is too thick, the object cannot be seen as clearly or may not be seen at all. Peeling, scraping, slicing and squashing are all techniques used to obtain thin specimens.

Staining Specimens

When looking at plant and animal cells under microscopes, we often need to stain the tissue we are looking at in order to see the different parts of the cell. There are many types of stains which biologists use and each stain reacts with different substances in the specimen. For example, iodine reacts with starch and turns it a blue-black colour. Take care when using stains because they can stain you as well.

Activity 1: Looking at Euglena

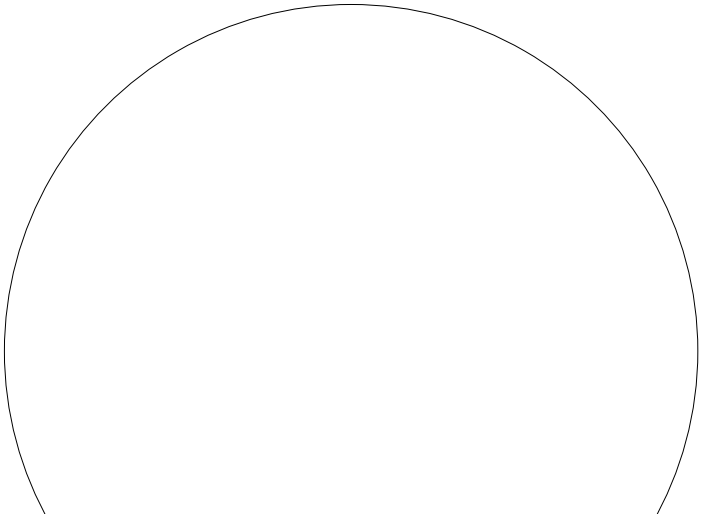
Euglena are unicellular organisms. They have chloroplasts and can make their own food, but they are also able to absorb food from their environment. They usually live in quiet ponds or puddles. Euglena move by a long whip-like structure, called a flagellum, which acts like a little motor and pulls the organism through the water.

Equipment needed: Microscope slide, cotton wool, Euglena specimens

Method: 1. Tease out the cotton wool and place one or two strands on your slide.

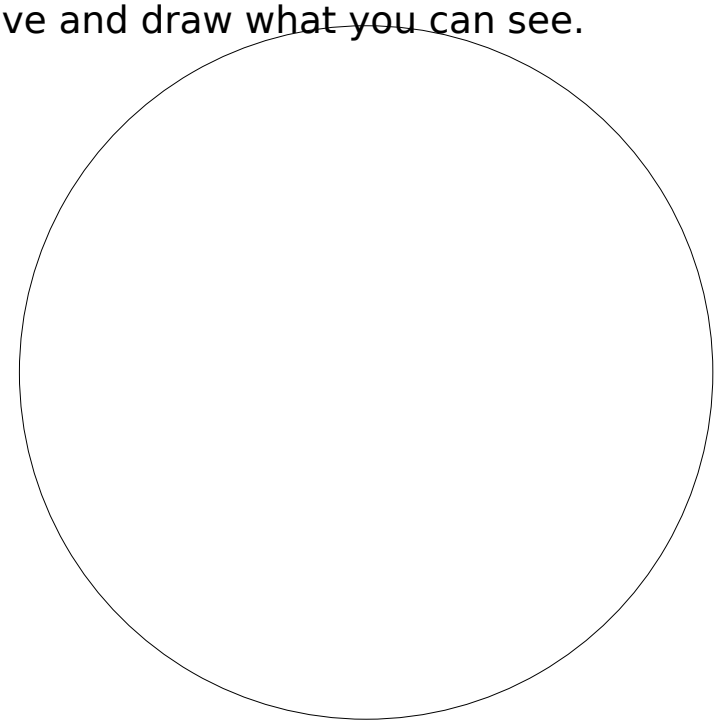
2. Put a drop of water containing the Euglena on the cotton wool.

3. Examine the slide under your microscope (under **low power** first) and draw what you see. The cotton wool should stop the Euglena swimming away.



Magnification =

4. Now look at the Euglena under the **medium power** objective and draw what you can see.



Magnification =

Activity 2: Looking at Onion Cells

An onion is made of layers, each separated by a thin skin or membrane. This membrane can be looked at by carefully peeling it off the onion and staining it. The membrane is on the rough surface of the onion layer.

Equipment needed: Microscope slide, coverslip, onion membrane, staining solution, toothpick, tweezers

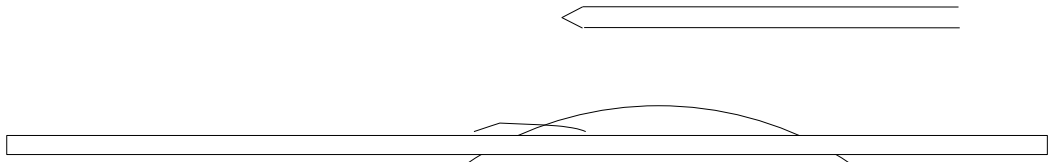
Method: 1. Take a small piece of onion and using the tweezers, carefully remove the membrane on the rough surface.

2. Place a small sample on the slide and add a drop of staining solution.

3. Carefully lower the coverslip using the toothpick.

4. Examine your slide under low power, medium power and high power. Draw and label what you can see.

How to lower a coverslip



Onion Cell Diagrams

Magnification =

Magnification =